

A Packing of 60 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 60 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.909433. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.908739, obtained by deleting 1 of the 61 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 60 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.909433 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 60$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.908739, obtained by deleting 1 of the 61 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 60/59 = 0.938722$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N60.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 69 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
909433456789497084483673258079878256143108580439
denominator:
1000000000000224568515536306293061353247987061400
```

B Coordinates

60 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N60.txt`.

338745432736	34508874299	47639492104	-735592998140
267068029839	144066532450	4101909350	186461114125
-132062871843	-25497280515	54761509591	-163417343619
119412382851	685598088775	-27358525557	-8842406783
-378301154190	404396761462	-592470242315	371978145961
-161509201087	-171074200612	40889837933	-125771012765
55151147428	-154149287889	198615600	-60501934007
-336278007458	-70643138590	330411569731	71341617658
190833893177	-357645918148	-144645509157	162883840496
117262612207	97103063440	301693071960	169762577189
-439268324116	72683817649	146726641941	197555109011
-393400924653	277223649195	-90372155081	250473446756
314667609050	137516536427	-269073273400	192227102266
479829569026	97954558282	341183404696	-10815027659
84013591684	-536436319196	-196758767910	-411119737899
-93409286188	264411094964	410157625357	-481648977189
77778597803	-436323448983	81418158032	-420014768618
-112509193160	132042846503	27421401605	73017526788
218540725247	-345133020154	-229451132127	-200783375424
-457293662556	-227934930612	473660014106	-286282882363
-302543766820	124325615505	-207388000742	94830719117
-160637171014	-60840258224	489534444547	-18719003157
-204234367781	-31131900958	-263099225932	36877581571
154367895588	176003175316	269569852666	223234549275
-502189298313	-178544090837	-186331198275	-69650326160
-208140308596	-358849138469	325778743045	153723407299
491359158149	509543931058	282073503217	-487312840028
276574981631	-95955174303	63972469109	-40746425911
152339834260	84548868447	-258003125924	23271965744
-119664124795	-4658417425	70759455790	-438046085804
-577903346056	387164340088	146658338500	84107170088
-775882179	-225669468682	276041125190	143293002063
-85013983197	-559421812605	213745458902	-319411546045
323195060007	-117953925480	510789067056	-47182252693
278931869375	192647201963	-41397829187	-617541838667
88554904631	-137118720140	70505194881	-102863001063
301689280521	387599282451	430592490190	-127131398352
-149083651079	-11260283266	-129548272356	-280211749292
-1705859980	10461075576	-321498116566	-306892233502
-357195737667	-277801691013	-437648530057	-504839540481
412194594157	-347955668330	223739078940	250578102667
225781333903	-45295625503	-191987696576	-353829552350
-332130098743	-122306534061	-454368362876	390279049718
-125140195471	364858059806	-252599192271	-159279174670
-53642056254	228887900919	-90472428836	-119650267456
321853858689	-270528128733	-641757058846	-112686199510
122100015134	565386618413	-230952489597	-258398394660
9224005939	-266790103111	-156368567968	-206461514398
-267239418228	-387122624786	250751774911	-204442685939
548491339380	-136942617245	-22565242076	168985135353
77988857976	-87180342461	-134584680153	-437368935151
160487847689	148320488766	105119558168	-201452180402
167854599910	-268631448855	32492085444	98989326890
-283217002017	-220818748293	-27549644279	-470215969302
292921071263	-216421141023	-370879974515	51401144529
-10789818802	-47703504111	-166268292397	344940599804
-52042422192	85493305390	-273350120887	-259505690214
-5677002525	-65306272448	-171790721239	-135257829307
40347220379	-738526384405	-222574325856	-326827699445
-313972070230	-60009069480	48393666889	-77963473949
-599833202775	58880339834	516153087694	167316833133
-185030370304	162359813149	-396619196769	-62373525195
-429427637269	155610740751	-375833247372	430324857019
-264165419284	-84814615380	158009822810	90991731135
233493499384	-48350161821	97430911039	-475553295799
-76373580623	566552060939	58839563150	-566294264692
-333995583519	-444335864854	110343017073	293715785144
65953281362	446412588790	-448522293509	111145122994
-330339546416	35568897023	116975229878	-121083365631
234400336002	-460626011718	355462902271	71130060927

-106302310493	262864951282	-312659257418	-222679398475
46181062345	132568627417	-151175911074	139125707959
-90970266165	-211359150736	364650969603	-107590142522
361139752346	204461982064	323380152309	95155684131
-55198342868	-654464626978	-291817497117	-238352518526
-413956841769	157251315315	-132271643072	-205277723917
140618986974	-398128636515	276576630252	-589272816522
194804228453	-181134725001	-562314079930	-465648703447
321468376278	-190410859938	211663461808	-49504584429
252037442680	-121663332537	163187649863	425686643699
-28596793922	-488142335846	-237532748749	137244025491
-624571353300	-96459376568	501463116823	-211684778042
-446537720920	-196504997525	-92938738275	21861949543
-200924968014	-157185304608	-208064462630	-84586967984
-106395375875	10783491920	-73631111040	2659320371
-116910779969	651850433375	-50423674728	585190034733
-329593645424	-68354827257	161736854082	-321224406990
91491855368	-349499134166	-438122904006	205166005230
-52421898802	149855693831	-479788798916	422008414769
-145919167142	183199632720	129464972921	226041460538
336927282144	173685980264	19709100206	181412401422
-57960707409	-87838624768	-407609857741	-379219864701
-84469081077	269730910323	-24843792985	303195364896
-145305404086	-293643923419	-169279091979	-45498328483
119987867059	167086596026	-456534480773	68796115768
-8923681522	-53211932561	-19102595186	-130907212381
-718245958873	101587167884	-283942196538	-13106072592
108564905832	-77926091170	-298887227241	64004093533
-414227368490	304312402373	-198115988541	126316073036
-7262698633	-48490694154	-257937540672	226864701511
333509563519	38667249734	233722897290	-51585202985
92593025574	369302859211	60665286167	-39524486649
501275561140	-649236697981	130674396474	-2758669193
-197159385478	363064120873	-514743199977	-362456754679
-366519942767	-514080903679	-14086569847	216478389275
-207277216190	-45761092330	382460483484	-123465696906
80850140831	145261683504	-307194878841	-176545616549
-570575537666	120234378382	-123135660517	-826090926574
-133369438646	-70628468125	171354152379	-96577220061
164941584781	129971181016	-87462868653	524125117187
303468853669	445378561032	-659645498488	365710166115
-9478131529	-258854880984	249187058995	312225524730
43044847300	212441490756	116358527504	-141871345857
-166193910125	79800931895	-43343522070	11454465014
183868153709	-63486414587	503749611654	26768410858
-35900849194	-392596807114	-24689345700	93495904888
-375752377545	448506778657	-128447536225	72139886853
390470249967	98537698580	254043960761	-303303241063
684300542288	-142572803171	-73499905605	-171859504996
114871336040	218030274722	162969453312	-519154325624
-109758718467	163246014035	154412651220	-242417738678
35600269043	-375653421244	-67810185931	319727640185
52476731154	295557807002	475341098381	-347948926347
-380811718078	-487950144356	155776770910	-38982077346
125820505983	-271116532850	306614618598	168818863816
88110130068	-184498532461	492774886090	222438702545
197105838648	136111337293	22201447000	506496573016
-7808579812	-144564877895	-306032411710	72642157492
414728596120	444392363414	276906767621	-231643949075
-293141608355	291132009277	-280143054820	-213381449357
377645989070	173575270321	-329557377186	-477889535891
64642638285	-78393459159	-5477135147	80672300651
-288570633338	-391647010438	320520574936	318161066421
20872592725	-61656159858	158309702877	9395575734
417315410086	438042873179	346299800126	79654434298
56390462955	336452520065	-591277306093	143409880975
98918778007	-61693952667	-105889770964	131753177537
441682377875	-477591946784	407582613427	-501103251414
-207789137431	182447113744	82470357944	-136100168340
169268669821	89639091970	-143488745416	-328509030418
239866333865	-16463969837	-283461747179	-252344875927
464450549154	-257094722127	-79965287905	309924356456
-39228723050	402323607634	115220174542	-368419666582
22411844546	116292810616	84669956399	-325574575979
-92461537795	-532809214476	-301257173966	34581327651
-360171447361	165121605383	-180380818441	-418312546820
-94819879620	281968011531	-527747445668	23233950345
88475613943	-141756833425	151142584169	-561116678314
124906996019	-218761875172	200177422176	7009196212
78033245470	-80145980314	-81439685685	350358165770
-661060308738	-565921574549	-178738690857	144456351758
-391829387621	354264915268	451357309224	170354713207
118476435986	-197875290960	131552795827	-270329690484
-29908876187	-3194792115	93790856093	123430314570
-176301134520	-29477989610	-157894310009	-296413410500
661784679526	188784096245	-458804325355	370286722314
-61331568859	315759786427	-202189083599	-80981684074
173505282357	104988531478	-285151959186	185644371855
450193266394	51996525831	-109680167063	64246356864
-152028901425	-69377208851	282087937112	-392458817208
208475464380	465959075176	-230592252303	-510009516546
-199796014403	-31342620006	-131701859932	-116032211778
276013003370	-246322956520	232923932206	316271582884
120354695173	-244157997707	273171332635	224992650235
-280454144699	555352033534	340884622986	-119933192811
31252302117	-382080300421	79989051047	-99107322727
132764485560	-69989555207	-240429369103	-119120026354
-172813523322	29205432473	-551251733765	64316485773
-235947053404	-288470537772	-215467620637	-12177339658
135167664344	-120078784850	-158759786267	-605567021179
-258724356902	27013728919	391262136589	-177255091568
-108341790473	-244785003207	9972725861	185826118830

316914136983	-414079253367	-198404107553	263013237205
-117929105628	177809177455	-150262178414	-579630635625
445196103608	-213163343374	-148412636124	135109569019
287809381460	-93075588131	106656906103	280359154939
483895870638	58562794046	107224449428	-458696228274
-389528504561	138420585878	531020095675	-257016027436
-168663214133	227722715282	-209267606076	9896288895
59175343175	-24594329353	-294217743580	-56656354045
-284940589150	322168606061	209404627511	87765286729
56855635621	71364540409	91538456112	-365941670965
264865521653	408615833009	127117579188	-39987396132
291248464237	-345989869980	-84832256634	152028737696
115454834906	-355747943696	134755196200	-483704538504
185712979521	169567603487	-607357581699	-9286010528
-165531029193	179941842095	-234573451262	204622904081
116713645146	276571024857	528602574644	143603637315
-234482290597	514981414295	31784702934	-416464591162
104748788657	-540592165508	-22404090099	58030054953
-455624141884	-31241200559	-258776179566	189629297729
-612776804039	-356048629447	218828789413	-150636720337
112797508122	199273324442	238338159611	-409977625052
-186331951648	-280227097613	-117154308380	476467920217
72884141500	-170760397989	-77992963274	219488395820